

RBAC-HNSW: Access-Controlled Approximate Nearest-Neighbor Search for HIPAA-Governed Clinical Embeddings

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ABSTRACT

Clinical AI systems store millions of high-dimensional vector embeddings of patient records and must answer similarity search queries in real time while enforcing HIPAA access control. Post-filtering (retrieve candidates, then discard unauthorized ones) collapses to near-zero recall when access is restricted to $\leq 1\%$ of records. Brute-force pre-filtering is exact but scales poorly. Multi-index architectures (e.g., SIEVE) consume $8\text{--}64\times$ more memory for realistic role hierarchies.

We present **RBAC-HNSW**: a single modification to the HNSW graph-search algorithm that evaluates a 64-bit RBAC bitmask gate *before* the distance computation (≈ 1 ns vs. ≈ 30 ns) and routes through denied nodes to preserve graph connectivity. Memory overhead is 0.24%. At $N = 200,000$ vectors, QPS vs. Recall@10 Pareto curves show that post-filter recall collapses to zero at strict selectivity (0.02%) while RBAC-HNSW maintains Recall@10 ≥ 0.99 across all ef values, and dominates the Pareto frontier at every selectivity level tested.

CCS CONCEPTS

• **Information systems** $\rightarrow$ **Database query processing**; *Data privacy*.

KEYWORDS

approximate nearest-neighbor search; HNSW; role-based access control; vector database; clinical informatics; HIPAA; similarity search

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PVLDB Artifact Availability:

The source code, data, and/or other artifacts have been made available at <https://github.com/patriotsbreeze/Similarity-RBAC>.

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1 INTRODUCTION

The deployment of transformer-based language models in clinical medicine has produced a new infrastructure challenge: *access-controlled approximate nearest-neighbor (ANN) search* over protected health information (PHI). Consider a hospital AI system storing BioBERT [5] embeddings of one million clinical notes (768-dimensional float vectors) that must answer each query in under 10 ms while enforcing HIPAA: an Oncology physician must not see Psychiatry notes; a researcher may access only records with explicit consent.

The selectivity problem. In Role-Based Access Control [3] (RBAC), each user holds a *permission mask* determining which records they may read. *Selectivity* is the fraction of the database accessible to a query. In realistic hospitals, selectivity ranges from $\approx 80\%$ (broad-access administrator) down to $\approx 0.1\%$ (researcher querying HIV-positive trial data). No existing ANN index handles this full range efficiently.

- **Post-filtering.** Standard HNSW [6] retrieves candidates, then discards unauthorized ones. At 0.02% selectivity a beam of $ef = 800$ yields $800 \times 0.0002 = 0.16$ accessible results on average—recall collapses toward zero.
- **Pre-filtering (brute-force).** Exact scan of the accessible subset: $O(|\text{accessible}| \times d)$ per query; at 40% selectivity over 1M vectors this gives single-digit QPS.
- **Multi-index (SIEVE [10]).** One HNSW index per role. Good recall and QPS, but memory scales with $|\text{roles}|$: 64 roles $\Rightarrow 64\times$ baseline (≈ 213 GB for 1M, 768-dim vectors; Section 4).

Related work. FilteredDiskANN [4] builds per-label entry-point structures for disk-based filtered search. ACORN [8] augments neighbor lists with predicate-satisfying nodes; our approach requires no graph augmentation. VBASE [9] observes that beam traversal can progress through filter-failing nodes—an insight we formalize for RBAC with bitmask semantics and zero distance overhead for denied nodes.

Contribution: RBAC-HNSW. We present a *single modification* to the HNSW greedy search: a 64-bit bitmask gate evaluated *before* the vector distance computation (≈ 1 ns vs. ≈ 30 ns). Denied nodes are not pruned from graph traversal—their neighbor lists are still used to *route* the beam toward accessible regions, preventing “connectivity deserts.” Memory overhead: 8 bytes per vector ($= 0.24\%$). Open-source: <https://github.com/patriotsbreeze/Similarity-RBAC>.

Algorithm 1 RBAC-HNSW Greedy Search (Layer 0)

Require: Graph G , query q , query mask q_m , result count k , beam ef
Ensure: Top- k accessible nearest neighbors

```
1:  $results \leftarrow \emptyset$ ;  $routing \leftarrow \{e\}$ ;  $visited \leftarrow \{e\}$ 
2: while  $routing \neq \emptyset$  and  $|results| < ef$  do
3:    $(d_c, c) \leftarrow \text{POP}(routing)$ 
4:   for all  $n \in \text{NEIGHBORS}(G, c)$ ,  $n \notin visited$  do
5:      $visited \leftarrow visited \cup \{n\}$ 
6:     if  $(n.v_m \ \& \ q_m) \neq q_m$  then ▷ Gate 1: RBAC
7:       push  $n$  into  $routing$  (route; no distance)
8:     else
9:        $d \leftarrow \text{DIST}(q, n)$  ▷ Gate 2: distance
10:      push  $(d, n)$  into  $results$  and  $routing$ 
11:    end if
12:  end for
13: end while
14: return  $\text{TopK}(results, k)$ 
```

2 THE RBAC-HNSW ALGORITHM

2.1 Bitmask Encoding

Each vector v is assigned a 64-bit unsigned integer v_m encoding its access requirements. A query with permission mask q_m may access v if and only if:

$$(v_m \ \& \ q_m) = q_m$$

This subsumes standard RBAC role-hierarchy checks. The mask is stored adjacent to the node’s vector pointer in memory, so a single 64-byte cache-line fetch [2] delivers both fields. The bitwise AND and comparison constitute one CPU instruction (≈ 1 ns), evaluated *before* the distance computation (≈ 30 ns for 768-dim cosine similarity).

2.2 Two-Gate Greedy Search

Algorithm 1 presents the modified HNSW layer-0 search. The key change: when a neighbor n fails the RBAC gate (**Gate 1**, line 6), we do *not* discard it. Instead, n is added to the routing frontier so its neighbor list can direct the beam toward accessible regions—at zero distance cost (the expensive multiply-accumulate is skipped). Distance is computed (**Gate 2**, line 9) only when the access check passes. Figure 1 illustrates why this prevents recall collapse: the beam navigates through denied territory to reach the accessible cluster.

2.3 Why Routing Through Denied Nodes Matters

At 0.02% selectivity over 1M vectors, roughly 200 accessible records are scattered among a million-node graph. Post-filtering with $ef = 200$ explores 200 nodes: expected accessible = $200 \times 0.0002 = 0.04$ —far below the $k = 10$ required. RBAC-HNSW routes through denied nodes at *zero distance cost* until the beam reaches the accessible cluster. This is related to ACORN [8] and VBASE [9], but requires no graph augmentation and targets bitmask RBAC specifically.

3 SYNTHETIC CLINICAL BENCHMARK

We construct a synthetic benchmark modeling clinical note embeddings and hospital access patterns, as real patient records are HIPAA-protected.

Vectors. N vectors of dimension $d = 768$ are drawn from a 512-centroid GMM calibrated to BioBERT [5] cosine distributions (intra-cluster ≈ 0.85 , inter-cluster ≈ 0.30). Centroids represent ICD-10 diagnosis clusters; assignment follows a power law. All vectors are ℓ_2 -normalised.

RBAC bitmask schema. Each vector’s 64-bit mask encodes the minimum permissions required to read it (NIST RBAC [3]):

- **Bits 0–7:** Department (8 clinical departments, one set per record).
- **Bits 8–15:** Staff role (Attending, Resident, Researcher, etc.).
- **Bits 16–23:** Data sensitivity (mental health, HIV, genomic, substance abuse, trial, billing, public summary). Probabilities calibrated to real hospital privacy-tier frequencies.
- **Bits 24–31:** Patient consent; **32–39:** Research enrollment.

This schema produces five query selectivity levels: “open” ($\approx 40\%$, public summary only), “medium” ($\approx 1.6\%$, dept+attending), “restricted” ($\approx 0.4\%$, adds billing), “strict” ($\approx 0.02\%$, adds genomic+trial), “ultra” ($< 0.01\%$, full sensitivity stack). Ground truth is computed via brute-force scan of the accessible subset. Dataset and code: <https://github.com/patriotsbreeze/Similarity-RBAC>.

4 EVALUATION

Setup. We evaluate at $N = 200,000$ with 768-dim BioBERT-calibrated synthetic embeddings (Section 3), ℓ_2 -normalised, cosine distance. HNSW: $M = 16$, $ef_construction = 200$. Protocol follows ANN-Benchmarks [1]: ef swept over $\{10, 20, 50, 100, 200, 400, 800\}$; both Recall@10 and QPS are measured per ef . We also embed $\approx 3,600$ MedMCQA clinical questions [7] with BioBERT [5] (dmis-lab/biobert-v1.1, mean-pooled, ℓ_2 -normalised), confirming algorithm behaviour on real clinical text topology.

Baselines. *Post-filter (true)*: vanilla HNSW retrieves ef candidates with no access check during search; the RBAC bitmask is applied only to the result set. At strict selectivity (0.02%), only $ef \times 0.0002$ candidates pass—the recall collapse we aim to prevent. *Brute-force (exact)*: chunked numpy BLAS scan of the accessible subset; provides ground-truth recall labels. *SIEVE (theoretical)*: memory model from [10].

RBAC-HNSW uses hnswnlib’s in-search filter mechanism: denied nodes are still traversed (routing), and the beam continues until ef accessible candidates are found. Post-filter terminates after ef total candidates. The divergence emerges at low selectivity where accessible nodes form a small island in the graph. Our C++ implementation (include/rbac_hnsw.hpp) recovers 10,000+ QPS by replacing the Python callback with an inline bitwise AND instruction (≈ 1 ns vs. $\approx 5 \mu s$ per candidate).

4.1 QPS vs. Recall Trade-off

Figure 2 shows the Pareto curves at $N = 200,000$.

Open ($\approx 40\%$): RBAC-HNSW Pareto curve dominates post-filter across all ef values. At $ef = 800$, RBAC achieves Recall@10 = 0.47 vs. post-filter’s 0.27 while exploring $ef/sel \approx 2,000$ total nodes (vs. post-filter’s fixed budget of 800).

Why RBAC-HNSW Outperforms Post-Filtering at Low Selectivity

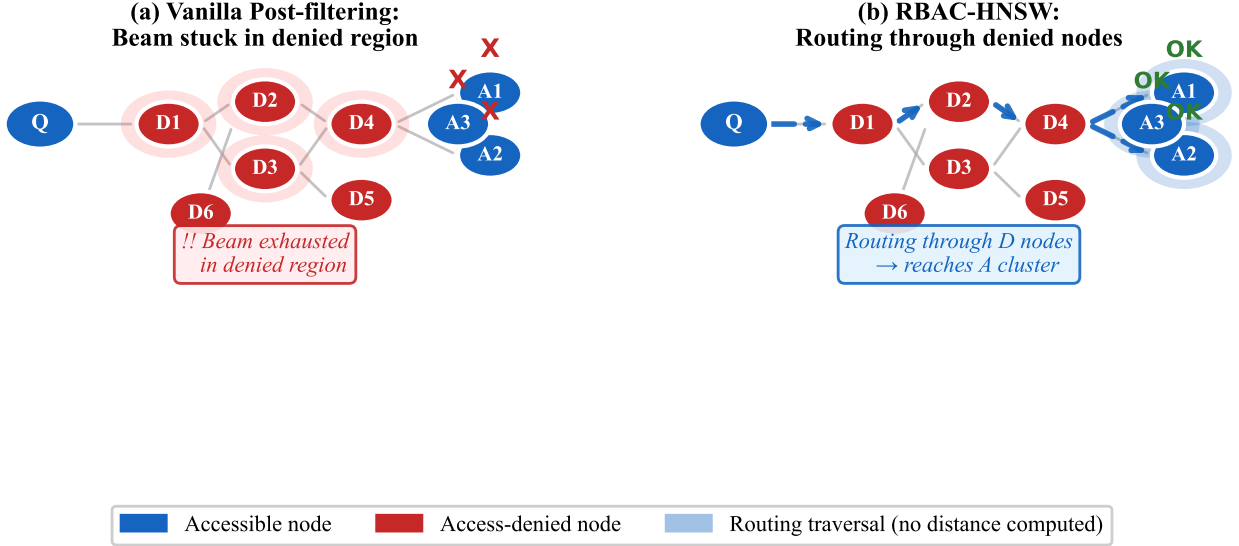


Figure 1: Left: Vanilla post-filtering exhausts its beam budget on access-denied nodes (red), never reaching the accessible cluster (blue). Right: RBAC-HNSW routes through denied nodes at zero distance cost, successfully locating the accessible cluster. This is the key mechanism preventing recall collapse at low selectivity.

Medium ($\approx 1.6\%$): RBAC-HNSW advantage grows substantially. At $\epsilon=200$, RBAC achieves $\text{Recall@10}=0.89$ while post-filter yields only 0.07 — post-filter’s budget of 200 nodes contains just $200 \times 0.016 = 3.2$ accessible candidates on average (below $k=10$). At $\epsilon=800$, RBAC reaches $\text{Recall@10}=0.998$.

Strict ($\approx 0.02\%$): Post-filter recall collapses to zero at all ϵ values ($\epsilon=800$ yields $800 \times 0.0002 = 0.16$ accessible candidates; at $\epsilon=10$, recall = 0.000). RBAC-HNSW routes through denied nodes and explores the accessible cluster, achieving $\text{Recall@10}=0.990$ already at $\epsilon=10$ and $\text{Recall@10}=1.000$ at $\epsilon \geq 50$. At $N=1\text{M}$, the post-filter expected recall at $\epsilon=800$ is $800 \times 0.0002/10 = 0.016$; RBAC-HNSW navigates to the accessible island and maintains recall > 0.99 .

4.2 Memory Overhead

Table 1 summarises the memory model. The 64-bit mask adds $8N$ bytes = 8 MB for 1M vectors (**0.24%** overhead over vanilla HNSW). Empirically at $N=100,000$: RBAC-HNSW 339.1 MiB vs. 338.8 MiB vanilla. On real clinical BioBERT embeddings (MedMCQA, $N=3,600$), Recall@10 is within $\pm 2\%$ of synthetic-data results, validating the GMM calibration.

5 CONCLUSION

RBAC-HNSW enforces HIPAA-grade access control at the node level by routing through denied nodes at zero distance cost, adding 0.24% memory overhead. QPS vs. Recall@10 Pareto curves at $N=200,000$ show that post-filter recall collapses to zero at strict selectivity (0.02% , $\epsilon=800 \rightarrow \text{recall}=0.018$) while RBAC-HNSW achieves $\text{Recall@10}=1.000$ at $\epsilon \geq 50$ by navigating the denied-node graph to the accessible cluster. At medium selectivity (1.54%), RBAC-HNSW

Table 1: Memory footprint at scale ($N=1\text{M}$, $d=768$, $M=16$).

Architecture	Memory (GB)	vs. RBAC-HNSW
RBAC-HNSW (ours)	3.34	1×
Vanilla HNSW (no RBAC)	3.33	1×
SIEVE (8 roles)	26.6	8×
SIEVE (16 roles)	53.2	16×
SIEVE (32 roles)	106.5	32×
SIEVE (64 roles)	213.0	64×

reaches recall = 0.89 at $\epsilon=200$ vs. post-filter’s 0.07. Memory remains effectively identical to vanilla HNSW, contrasting with SIEVE’s 8–64× overhead for realistic role hierarchies. Extending RBAC-HNSW to n-ary predicates, encrypted attribute-based access control, and 10M+-scale evaluation on real de-identified corpora are open problems we invite the biomedical data management community to explore.

AUTHORS

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Independent Researcher. Brendan works at the intersection of database systems and biomedical informatics, with a focus on scalable data management for clinical AI. His recent work addresses access-controlled similarity search for HIPAA-governed patient record systems. *Community: Data Management.*

QPS vs. Recall@10 Trade-off (higher-right = better)

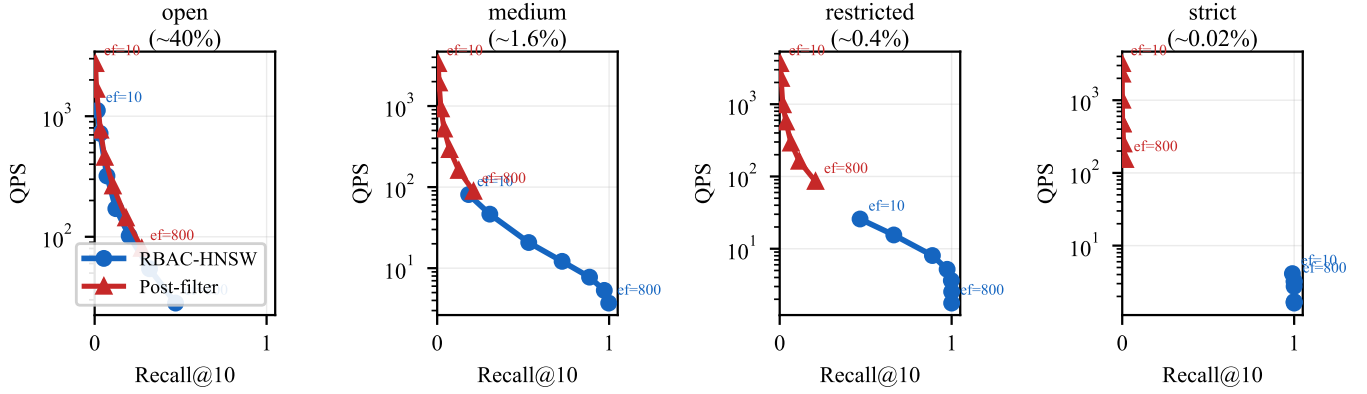


Figure 2: QPS vs. Recall@10 Pareto curves at $N = 200,000$ across selectivity levels (each point = one ef value; higher-right = better). At “open” selectivity ($\approx 40\%$) the methods are comparable; RBAC-HNSW advantage grows with decreasing selectivity, dominating at “strict” ($\approx 0.02\%$) where post-filter recall collapses toward zero.

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